



Real Decentralization

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A testable rubric separating distributed control from theater.

1 The distinction is about powers, not appearances

Decentralization is a property of powers, not of appearances. Examined adversarially, every system reduces to a small set of capabilities that can harm or halt its users: the power to change the code, to move the funds, to stop the system or exclude a participant, and to set the rules that govern the first three. A system is decentralized to the exact extent that each such power either requires the independent consent of many parties or has been removed entirely. It is centralized at every power that one party, or a small group able to collude, can exercise alone.

Decentralization theater is the substitution of the appearance of distribution for the distribution of the powers themselves. A validator count, a token ticker, a governance forum, and an off-chain poll are all appearances; none of them constrains a private key that can upgrade a contract or move a treasury. A rubric separating real from theater therefore ignores appearances and enumerates the powers.

The compact form is the walk-away test: suppose the founding team becomes adversarial tonight, what can it do to users before anyone reacts, and what can users do without it.

The quantity to measure, for any single power, is the size of its effective control set: the minimum number of independent parties whose collusion suffices to exercise it. This is the Nakamoto coefficient, generalized past consensus. It applies to a bridge multisig, an upgrade key, a governance vote, and the RPC providers a front end depends on. Two systems can display the same nominal count and differ entirely in the effective one: what matters is not how many signers a multisig lists but how many independent people must actually agree. Ronin listed nine validators and had an effective control set of one [public record].

2 The rubric

Score each power from the state a hostile analyst can demonstrate. **0 (concentrated)**: one party, or a small colludable set, exercises it with no delay and no outside consent. **1 (constrained)**: still held by few, but bounded by a timelock longer than the time needed to exit together with a unilateral user exit (forced withdrawal, freeze, or fork), honest training wheels. **2 (distributed or removed)**: the independent consent of many parties, or the power removed by immutability.

Power	Theater (0)	Genuine (2)
Upgrade	Instant proxy swap by a team multisig; no timelock	Immutable, or upgrade only via a broad on-chain vote plus a timelock past exit time
Custody	onlyOwner mint / sweep / rescue; a multisig holds the locked assets	No admin can move principal; mint capped or removed
Halt / censor	Single sequencer or whitelisted set; owner pause; no forced inclusion	Permissionless production, or a bounded pause with an escape hatch
Governance	Off-chain poll a multisig implements at will; insiders alone meet quorum	Governor to timelock to execution none can override; no insider majority
Exit	Closed source; balances are operator IOUs; withdrawal stops if the operator does	Open, forkable code; self-custody; unilateral forced withdrawal
Liveness	Halts if one firm's front end, RPC, oracle, or indexer stops	Contracts callable directly; many front ends and RPCs; decentralized oracle

Report the six scores as a vector, not a single total: a scalar hides the fact that matters most, which power is centralized, and the real ceiling is the lowest score among the powers load-bearing for users' funds and liveness. One instant-upgrade key returns the whole system to zero on the axis it controls, whatever the other scores say. A low score is not by itself theater: a system centralized and honest about it is immature, not dishonest. Theater is the gap between the claimed state

and the measured one, which the rubric records in an honesty flag. A zero disclosed is training wheels; a zero marketed as decentralization, or as “secured by” the settlement chain, is the case the rubric exists to catch. Every cell is checkable by someone who does not trust the project, tracing to an address, a function selector, a stake distribution, or a public incident. The L2BEAT Stages framework applies this discipline to rollups [public: L2BEAT methodology]; the rubric generalizes it.

3 Both sides, on the record

Genuine (on one axis does not imply genuine on all). Uniswap v2 core pairs are non-upgradeable with no admin power over deposited liquidity, and SushiSwap’s 2020 fork of that open code proved exit. Ethereum L1 carries no administrative key, staking is permissionless, and its concentration is measured and published rather than hidden. MakerDAO changes execute only after a governance timelock MKR holders control. Strongest of all is a removed power: Tornado Cash’s immutable contracts kept running after their deployers were sanctioned, because no key existed to stop them [public record].

Theater. Ronin listed nine validators at five-of-nine with five keys reachable through one organization, and lost about \$625M to that one compromise. Multichain marketed distributed MPC while one operator held the keys; when it was detained, roughly a billion dollars moved unrecoverably. Harmony’s Horizon bridge was two-of-five: two keys moved everything. Signaling governance, an off-chain poll a multisig implements at its discretion, is voice without power; the test is whether execution is automatic and un-overridable or manual and discretionary. A Stage-0 rollup sold as settlement-grade pairs a single sequencer with an instant upgrade multisig that can rewrite state and reach bridged funds, where “secured by” describes data availability, not control [public: L2BEAT methodology].

4 Applying it: Morpheus

Scoring Morpheus uses only published facts. The owner of the Base compute diamond `0x6aBE...030a` and of the capital contracts is a five-of-nine Gnosis Safe, `0x1FE0...eBa1E` [verified: on-chain].

Each of three funded powers is one `onlyOwner` call. **Upgrade 0:** the funded contracts are UUPS proxies whose `_authorizeUpgrade` is gated only by `onlyOwner` (`DistributionV6.sol:776`), upgradeable in one block with no timelock. **Custody 0:** the owner adds itself as minter and mints with no supply ceiling (`MOROFT.sol:50,61`), sets any user’s staked balance through `manageUsersInPrivatePool` (`DistributionV6.sol:175`), and about \$20M in staked stETH sits behind the same Safe. **Governance 0:** `MOROFT` is a LayerZero OFT, not `ERC20Votes`, holding no checkpoints (`MOROFT.sol:35`); the MRC process is Markdown proposals with no on-chain execution, last touched November 2025, so rule-setting is the owner call `editPool` / `editPoolLimits` (`DistributionV6.sol:102,118`) directing the roughly 76% of issuance in private pools [verified: repo]. The full vector is (0,0,0,0,1,1): Halt is a fourth zero, on a single-sequencer Base diamond the same owner can recut; Exit and Liveness reach 1 because MOR is self-custodied and callable through several front ends, but not 2 because the same upgrade key still reaches staked principal.

The July 2026 stipend-leak fix, applied to live Base within a day [public record], is direct evidence of the upgrade power: the same key that closes a bug can open one. Whether this reads as theater or as honest training wheels turns on the honesty flag, and the advisory MRC process, presented as governance while binding nothing, is the clearest theater on the vector. The path from 0 to 1 is specified in ADOPT-03: transfer `owner()` to a timelock the token controls, so every mint, upgrade, and pool edit becomes propose, vote, delay, execute [see ADOPT-03]. The path from 1 to 2 is distributing the vote itself, which thin participation can still defeat.

5 The mirror: our own stack, held to the same test

A migration passes only if the destination clears the rubric, and deploying a governance contract is not by itself a passing score. Three failure modes would reproduce Morpheus’s zeros under a nicer name: a launch multisig whose signers derive from one seed (effective control set one, whatever the threshold reads); a validator set one operator runs (a high replica count in one cluster is a

Nakamoto coefficient of one); and a vote whose quorum a single emission recipient can meet. The table scores the components we would actually hand Morpheus, read from our own repositories at their current state, not their design ceiling.

Component, as it stands today	Up	Cus	Halt	Gov	Exit	Live	Flag
Morpheus (Base diamond + capital)	0	0	0	0	1	1	open
Lux primary network / Quasar	2	2	1	2	2	1	disclosed
L2 from 12-template, at birth	0	0	0	0	2	0	scaffold
lux/dao governance, as deployed	0	0	–	0	2	1	disclosed
lux/ai node, served path	–	–	0	–	2	0	disclosed
mord settlement, PoC	0	0	0	–	2	0	clean

“–” marks a power the component does not bear. The Lux primary network penalizes misbehavior by reward forfeiture, not principal slashing (`uptime_forfeiture_mainnet_test.go:159`), and its mainnet genesis seats five initial stakers and five bootstrappers (`configs/mainnet/pchain.json`), so its Halt and Governance marks read the protocol’s permissionlessness, not a measured live set.

lux/dao governance. The mechanism is real: `ModuleGovernorV1` executes through a Safe, `VotesERC20StakedV1` is a non-transferable `ERC20Votes` stake, a freeze guard lets holders freeze a rogue Safe. What it sits on is the finding: `governance.md` records all five launch signers derived from one `LUX_MNEMONIC` seed, so the 3-of-5 has an effective control set of one until a rotation the document lists as unscheduled, and the timelock path is deployed beside the Safe but authoritative only “once token distribution lands and quorum is reachable” [verified: `lux/dao/governance.md`]. The reference instance `zoo/vote` ships with `moduleGovernor` and `governor` set to `null` [verified: `zoo/vote/config/zoo.json`]. Every module is `UUPS Ownable2Step`, so decentralization is as strong as who owns the proxies, today the single-seed Safe.

12-template and the validator set. The generated node manifest runs a five-replica `StatefulSet` in one operator’s cluster (`init.sh:240`) with single-replica MPC, KMS, and bridge, and the generated `Token.sol` grants one `admin` both roles with an uncapped mint (`init.sh:181,184`) [verified: `lux/l2-template`]. The software supports permissionless validation (`AddPermissionlessValidatorTx`, a `μLUX` minimum, `config/flags.go:320`), but the template does not distribute the set: a chain born from it is centrally sequenced at birth, decentralizing only if independent validators later stake in.

lux/ai, the heal-thyself finding. `cmd/lux-ai/main.go` mounts every route through `corsMiddleware` alone (line 262); no route verifies the `Authorization` header it lists in `CORS`, and `handleChatCompletions` returns a fixed placeholder (tokens 10/20/30, lines 279, 327), never calling the quorum settlement `pkg/rewards` names the canonical A-Chain path (`rewards.go:382`) [verified: `lux/ai`]. The quorum-and-slash engine exists; the served endpoint does not invoke it. Our public inference path is, as shipped, unauthenticated and not quorum-settled.

mord. A single-operator proof of concept that says so: “not deployed anywhere and it is not audited” [verified: `lux/mord`]. Its settlement improves on Morpheus, a consumer countersignature settling inside block acceptance, but that is integrity, not decentralization: an honest zero.

Verdict. As the repositories stand, swapping the five-of-nine for `lux/dao` plus a Lux L2 would, with today’s defaults, reproduce an effective control set of one under a better name, the exact substitution the migration must avoid. Four checkable changes close the gap: rotate the launch Safe to independent hardware keys held by distinct people; move `owner()` of the modules and `admin` roles to the token-controlled timelock and fill the null governor; spread the L2 validators, MPC, KMS, and bridge across independent operators; and route the served RPC through authentication and the quorum engine. The last we can do today; the validator set and a vote with non-insider participation our software makes possible but cannot make true, since thin participation defeats the vote and an empty validator market the sequencer. Real decentralization is those changes measured, not the mechanisms shipped.

RUBRIC. Method behind ADOPT-03 and LP-318. • research@lux.network